

C1 · Parsing-Retrieval Disconnect

Candidate parsers



Qwen3-VL



MinerU



Marker



PaddleOCR

Cleaner-looking ≠ better retrieval

parser	Boundary Clarity	Hit@1
MinerU	0.72 (highest)	0.20
Prod (ours)	0.61	0.55

parser choice alone moves Hit@1 by **+35.1pp**

(0.20→0.55; 2.8×)

appearance metrics measure formatting, not content

(anti-correlation $r = -0.81$, $n = 5$)

C3 · RCPS Selection (+ C2 Diagnose)

C3 · RCPS selection protocol (no training)

held-out Q-A probe, scored over 3 retrievers:



BGE-M3



mE5



Qwen3-Emb

extrinsic · retriever-averaged · format-normalised

no training · no manual chunk labels

→ **rank parsers & chunkers by retrieval**

ranks Prod far above MinerU (0.583 vs 0.212)

▼ **score low? locate which layer is at fault** ▼

C2 · Coverage diagnostic (no retriever)



covered

absent

absent = **parser-side failure**, set before chunking

no chunker can recover it → fix the parser (C4)

C4 · Bounded Parser Training

C4 · Parser-side training

best-of-K parses from Prod, ranked two ways:

RADP-Distill — edit-distance → ref
(reward-free control)

RADP-DPO — page-local RCPS
(retrieval reward)

Result

Hit@5 gain over the untuned parser:

RADP-Distill  **+1.22 pp**

RADP-DPO  **+0.85 pp**

substantially overlapping CIs
no evidence retrieval reward beats fidelity control